

PRACTISE QUESTIONS

Topic : Inheritance, Polymorphism, Operator Overloading

(a) Write a program to demonstrate runtime polymorphism.

Instructions:

1. Create a base class Shape with a pure virtual function calculateArea().
2. Derive two classes:
 1. Circle (with a radius as a data member).
 2. Rectangle (with length and width as data members).
3. Implement the calculateArea() method in both derived classes to calculate the area of the respective shapes.
4. Write a main() function to:
 1. Dynamically create objects for both derived classes.
 2. Call their calculateArea() methods using a base class pointer.

(b) Write a program to demonstrate polymorphism using function overriding.

Instructions:

1. Create a base class Employee with a virtual function getSalary().
2. Derive two classes:
 1. FullTimeEmployee (with basic salary and bonus as data members).
 2. PartTimeEmployee (with hourly wage and hours worked as data members).
3. Implement the getSalary() method in both derived classes to calculate the salary for each type of employee.
4. Write a main() function to:
 1. Dynamically create objects for both derived classes.
 2. Call their getSalary() methods using a base class pointer.

(c) Write a program to demonstrate abstract classes.

Instructions:

1. Create an abstract class Animal with a pure virtual function makeSound().
2. Derive two classes:
 1. Dog (implementing makeSound() to display "Woof Woof").
 2. Cat (implementing makeSound() to display "Meow Meow").
3. Write a main() function to:
 1. Dynamically create objects for both derived classes.
 2. Call their makeSound() methods using a base class pointer.

(d) Write a program to overload the + operator.

Instructions:

1. Create a class Complex to represent a complex number (with real and imaginary parts).
2. Overload the + operator to add two complex numbers.
3. Write a main() function to:
 1. Create two Complex objects.
 2. Use the + operator to add them and display the result.

e) Write a program to overload the () operator.

Instructions:

1. Create a class Functor with an integer data member.
2. Overload the () operator to allow the object to act as a function, multiplying the stored integer by a given value.
3. Write a main() function to demonstrate the functor behavior.

f) Inventory Management System

Requirements:

Create a Product class with the following:

1. **Private members:** productID (int), name (string), quantity (int), price (float)
2. **Public methods:**
 1. Constructor to initialize the product details.
 2. addStock(int) to increase the quantity.
 3. sellStock(int) to decrease the quantity.
 4. displayProduct() to print product details.

Create an Inventory class to:

1. Maintain a collection of Product objects.
2. Provide methods to add a product, update stock (add or sell), and display all products.

Instructions:

- Implement both classes.
- Write a main() function that:
 - Adds at least three products to the inventory.
 - Updates stock (add or sell) based on user input.
 - Displays the details of all products.